

in crime forensics and cybersecurity, but also by compliance auditors, fraud investigators, and even social scientists.

Free and open source tools are democratising access to the same powerful analytical capabilities that were once only available to those in resource-rich organisations. This means the gap between these institutions and smaller organisations is closing quickly, and thorough, competent, digital investigations are within everyone's reach.

Here are a few important and popular open source tools used for cybersecurity and information extraction.

- **Wireshark** (<https://www.wireshark.org/>)

Wireshark is a powerful network protocol analyser that is deployed for capturing and analysing network traffic. This software provides a microscopic analysis of the network environment for investigating traffic.

Wireshark integrates deep inspection of hundreds of protocols, with more being added all the time. It supports both live capture and offline analysis, and presents a standard three-pane packet browser to aid intuitive navigation. It is cross-platform, running on Windows, Linux, macOS, FreeBSD, NetBSD, and many other systems. Network data can be explored via a graphical user interface (GUI) or the TTY-mode TShark utility. It has some of the most advanced display filters in the industry and advanced VoIP analysis capabilities.

It supports a variety of capture file formats, such as tcpdump (libpcap), Pcap NG, Cisco Secure IDS iplog, Microsoft Network Monitor, and others. Besides, it decompresses a capture file that was compressed by gzip in real time. Depending on the platform, live data can be read through a multitude of network technologies: Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, etc. It also



Figure 2: Wireshark network protocol analyser

supports protocols such as IPsec, ISAKMP, Kerberos, SNMPv3, SSL/TLS, WEP, and WPA/WPA2 for decryption. Users may apply colouring rules to the packet list for quick, intuitive analysis, and output can be exported to XML, PostScript, CSV, or plain text.

- **Nmap** (<https://nmap.org/>)

Nmap (Network Mapper) is a free and open source utility for network discovery and security auditing. It is used by many systems and network administrators for network inventory, service upgrade scheduling, and monitoring host or service uptime. Nmap uses raw IP packets in unique ways to determine what hosts are on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics. It was created for quickly scanning large networks

but works well against single hosts too. Nmap is available for all major computer operating systems, with official binary packages available for Linux, Windows, and macOS X. It includes the classic command-line Nmap executable, but also has one advanced GUI and results viewer (Zenmap); a flexible data transfer, redirection, and debugging tool (Ncat); a utility for comparing scan results (Ndiff); and a packet generation and response analysis tool (Nping).

- **Autopsy** (<https://www.autopsy.com/>)

Autopsy is an open source digital forensics platform that provides a full suite of applications for investigating and analysing digital evidence. It is commonly used in forensics analysis of hard drives, smartphones, and other digital devices in the context of criminal investigation, security audit, and data recovery.

Autopsy has its own full forensics toolset: file system

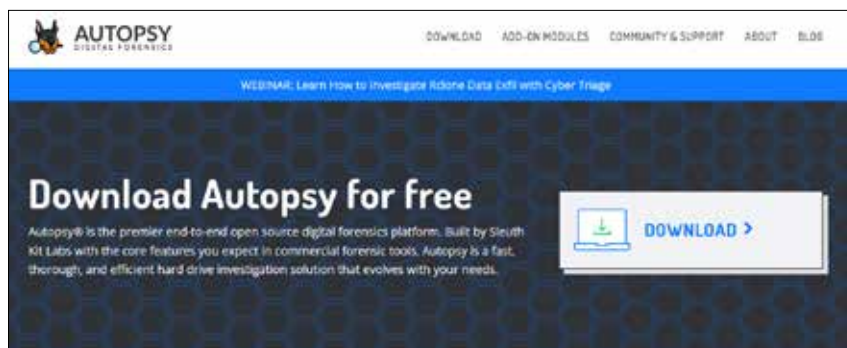


Figure 3: Autopsy tool for digital forensics